

MOUNIKA GERIKI

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PROFESSIONAL SUMMARY

Data Engineer & ML/AI Engineer with 3+ years of industry experience (TCS) and ongoing research at Stony Brook University. Specialized in building scalable ETL/ELT pipelines, cloud data platforms (AWS, GCP), and LLM-powered systems. Proven track record optimizing high-volume pipelines processing 50+ GB/day, fine-tuning large language models with LoRA, and architecting production-grade streaming and batch workflows. Seeking full-time Data Engineer or ML Engineer roles; authorized to work in the US on OPT with STEM extension.

TECHNICAL SKILLS

Cloud & Infrastructure

AWS: S3, EMR, Lambda, Kinesis, SNS/SQS, DynamoDB, Redshift, Glue, CloudWatch

GCP: BigQuery, Pub/Sub, Dataflow (Apache Beam), Cloud Composer, GCS

Azure: Azure Data Factory, Azure Databricks, ADLS Gen2 (familiar)

Data Engineering & Pipelines

Orchestration: Apache Airflow, Cloud Composer, dbt

Processing: Apache Spark, PySpark, Hadoop, Hive, Apache Beam, EMR

Streaming: Apache Kafka, AWS Kinesis, GCP Pub/Sub, Event-Driven Architecture

Patterns: ETL/ELT, Medallion Architecture (Bronze/Silver/Gold), Data Lakehouse, Dimensional Modeling, Star Schema, SCD

Quality: Data Validation, Reconciliation, SLA Management, Great Expectations (familiar)

Databases & Warehouses

Cloud Warehouses: Snowflake, Amazon Redshift, Google BigQuery

Relational: PostgreSQL, MySQL, Oracle (PL/SQL)

NoSQL: AWS DynamoDB, DuckDB

Languages & Scripting

Primary: Python (Pandas, NumPy, PySpark), SQL, HiveQL, SparkSQL

Secondary: PL/SQL, Java, GO(Golang), Scala (basic), Bash / Shell Scripting, C/C++, HTML, CSS

ML / AI & LLM Engineering

LLM & GenAI: LangChain, LangGraph, FAISS, RAG Pipelines, LoRA Fine-Tuning, Prompt Engineering

ML: Feature Engineering, LightGBM, BERT, Scikit-learn, Model Deployment, REST/SOAP APIs

Experimentation: A/B Testing, Evaluation Metrics, Inference Benchmarking

BI, Analytics & Reporting

Visualization: Power BI, Tableau, Looker Studio

Spreadsheets: Microsoft Excel (Advanced — pivot tables, VLOOKUP, Power Query, data modeling)

DevOps, Collaboration & Methodology

DevOps: Docker, Git, GitHub Actions, CI/CD Pipelines, Linux/Unix

Workflow: Agile / Scrum, Jira, Confluence, Cross-functional Stakeholder Collaboration

PROFESSIONAL EXPERIENCE

Graduate Research Assistant – Data & LLM | [Stony Brook University](#) | Stony Brook, NY Jan 2025 – Present

- Designed and maintained ML feature engineering pipelines for domain-specific LLMs (MaterialsBERT, BatteryLLM, PolymerLLM), processing structured and unstructured telemetry datasets at research scale.
- Applied LoRA fine-tuning and prompt-engineering techniques, improving structured signal extraction from unstructured scientific text by 20% and reducing model iteration cycles.
- Built automated evaluation and inference benchmarking workflows enabling reproducible LLM training pipelines with consistent data quality guarantees.

- Productionized training and inference pipelines in collaboration with ML researchers, applying software engineering best practices (versioning, validation, monitoring) to academic ML workflows.

Data Engineer | [Tata Consultancy Services \(TCS\)](#) | Bengaluru, India

Jun 2021 – Jan 2024

Event-Driven Banking & Customer Analytics Platform

- Engineered scalable ETL pipelines in Python and SQL, ingesting and processing 50+ GB/day of transactional banking data via REST/SOAP APIs into a centralized AWS data lake serving 200K+ customers.
- Built distributed PySpark transformations on AWS EMR, reducing end-to-end processing time by 45% and improving pipeline stability by 45% through workflow automation and optimized execution plans.
- Designed star-schema dimensional models in Snowflake and Redshift for enterprise reporting; optimized complex SQL queries reducing dashboard latency by 40% and improving reporting efficiency by 35%.
- Implemented event-driven ingestion using AWS Lambda and Kinesis, supporting near-real-time workloads alongside batch pipelines with 99.8% SLA adherence.
- Automated data validation and reconciliation frameworks, proactively catching schema drift and data quality issues before downstream consumption.

Snowflake Migration & Data Platform Modernization

- Led end-to-end migration of 50+ legacy Oracle reports and stored procedures to Snowflake, redesigning transformation logic for cloud-native performance.
- Built reusable ELT components standardizing ingestion from 5+ heterogeneous source systems; applied clustering keys and partitioning strategies to improve warehouse query performance.
- Partnered with BI and product teams to build curated data marts supporting 200+ daily business users, aligning data models to business KPIs (MRR, ARR, churn, retention).

Machine Learning Intern | [VI Solutions](#) | Bengaluru, India

Sep 2020 – Jan 2021

- Developed ML-driven data pipelines for healthcare analytics achieving 87% prediction accuracy and applied OCR and clustering pipelines to medical imaging workflows improving processing speed by 30%.
- Integrated real-time analytics pipelines with sub-500ms latency into production systems.

SELECTED PROJECTS

NYC Taxi Insights – Cloud Data Platform (GCP, BigQuery, Airflow, Looker Studio)

- Architected Airflow-orchestrated ELT pipeline on GCP Composer processing 9+ months of trip event data; designed star-schema models in BigQuery improving query performance by 35%.
- Implemented LightGBM and BERT-based ML forecasting pipelines achieving 0.012 MSE; built real-time dashboards for demand forecasting with monitoring and cost optimization.

TravelWise-NYC – Agentic RAG Pipeline (Python, GCP, FAISS, LangChain)

- Built event-driven retrieval pipeline with FAISS vector indexing and adaptive routing; developed evaluation metrics framework tracking accuracy, latency, and failure rates for ML inference reliability.

InsightPilot – Agentic AI Copilot (Python, SQL, DuckDB, LangGraph, Streamlit)

- Built an agentic platform integrating data ingestion, SQL analytics, and natural-language querying; designed feedback loops and latency/accuracy evaluation metrics for production readiness.

Patient Vital Monitoring System (GCP Pub/Sub, Dataflow/Beam, BigQuery, Power BI)

- Engineered real-time streaming pipelines to ingest, validate, and enrich patient vitals with automated alerting; implemented Bronze/Silver/Gold medallion architecture with data quality checks.

EDUCATION

M.S. Data Science | GPA: 3.58

Jan 2024 – Dec 2025

Stony Brook University, Stony Brook, NY

Relevant: Big Data Analysis, Statistical Computing, Data Management, Data Structures & Algorithms, LLMs

CERTIFICATIONS

Academy Accreditation – Databricks Fundamentals (Databricks, Feb 2026) · Google Cloud IAM and Networking for AWS Professionals (Google Cloud Security, Mar 2026) · Google Associate Cloud Engineer (In Progress) · GenAI in Data Analytics – Meta · Machine Learning with Python · 5-Day Generative AI Intensive – Google/Kaggle · Python for Everybody – Coursera · Python Data Structures – Coursera